

British Standard

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Steel fabric for the reinforcement of concrete

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Committees responsible for this British Standard

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- ISE/9 Steel for concrete reinforcement
- ISE/9 Bars, wire and fabric for concrete reinforcement

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Foreword

This British Standard has been prepared by Subcommittee ISE/9/1.

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Summary of pages

This document comprises a front cover, an inside front cover, pages i and ii, pages 1 to 8, an inside back cover and a back cover.

1 Scope

This British Standard specifies requirements for factory-made welded steel fabric manufactured from plain or deformed wires conforming to BS 4449 or BS 4482 for reinforcement of concrete.

2 References

The following normative documents contain provisions that, through reference in this text, constitute provisions of this British Standard. For dated references, subsequent amendments to, or revisions of, any of these publications do not apply. For undated references, the latest edition of the publication referred to applies.

BS 4449:1997, *Specification for carbon steel bars for the reinforcement of concrete*.

BS 4482:1985, *Specification for cold reduced steel wire for the reinforcement of concrete*.

BS 4483:1985, *Specification for steel fabric for the reinforcement of concrete*.

BS 4466:1989, *Specification for scheduling, dimensioning, bending and cutting of steel reinforcement for concrete*.

BS 8110-1:1997, *Structural use of concrete — Code of practice for design and construction*.

ISO 10287:1992, *Steel for the reinforcement of concrete — Determination of strength of joints in welded fabric*.

3 Definitions

For the purposes of this British Standard the following definitions apply.

3.1

batch

quantity of fabric of one type or steel grade presented for examination and test at one time

3.2

bundle

two or more sheets of fabric bound together

3.3

transverse wire

reinforcing element perpendicular to the manufacturing direction of the fabric

3.4

longitudinal wire

reinforcing element in the manufacturing direction of the fabric

3.5

welded fabric

arrangement of longitudinal and transverse bars or wires of the same or different diameter and length, arranged substantially at right angles to each other, and factory electrical resistance welded together by machine at the points of intersection

3.6

nominal size

diameter of a circle with an area equal to the cross-sectional area of the wire

3.7

length

(of sheet) longest side of a sheet of fabric, irrespective of the manufacturing direction

3.8

pitch

(of fabric) centre to centre distance of wires in a sheet of fabric

3.9

width

(of sheet) shortest side of a sheet of fabric, irrespective of the manufacturing direction

4 Information to be obtained from the purchaser

To assist in supplying the required fabric, the following information shall be obtained from the purchaser at the time of enquiry and/or order:

- a) the type of wire required and the number of the British Standard to which it conforms, i.e. BS 4449 or BS 4482;
- b) the required wire size, grade and mesh arrangement;
- c) the dimensions of each sheet required;
- d) the quantity of each sheet required.

5 Dimensions

5.1 The dimensions of the individual wires shall conform to the appropriate British Standard, i.e. BS 4449 and BS 4482, except for D49 wrapping fabric of 2.5 mm diameter.

5.2 The combination of mesh, wire size, wire grade and sheet dimensions for welded steel fabric shall be specifiable in accordance with annex B, or, for the preferred range of standard fabric types, shall be as specified in Table 1. The combination shall conform to the tolerances specified in clause 10.

6 Cross-sectional area and mass

6.1 The cross-sectional area and mass of an individual sheet shall be derived from the specified dimensions of the sheet, the nominal wire sizes and the specified pitches for the wires.

6.2 The cross-sectional area and mass per square metre of the preferred range of standard fabric types shall be as specified in Table 1.

6.3 The actual cross-sectional area and mass of welded steel fabric shall conform to the tolerances specified in clause 10.

Table 1 — Preferred range of designated fabric types and stock sheet size

Fabric reference	Logitudinal wires			Transverse wires			
	Nominal wire size mm	Pitch mm	Area mm ² /m	Nominal wire size mm	Pitch mm	Area mm ² /m	Mass kg/m ²
Square mesh:							
A393	10	200	393	10	200	393	6.16
A252	8	200	252	8	200	252	3.95
A193	7	200	193	7	200	193	3.02
A142	6	200	142	6	200	142	2.22
A98	5	200	98	5	200	98	1.54
Structural mesh:							
B1131	12	100	1131	8	200	252	10.9
B785	10	100	785	8	200	252	8.14
B503	8	100	503	8	200	252	5.93
B385	7	100	385	7	200	193	4.53
B283	6	100	283	7	200	193	3.73
B196	5	100	196	7	200	193	3.05
Long mesh:							
C785	10	100	785	6	400	70.8	6.72
C636	9	100	636	6	400	70.8	5.55
C503	8	100	503	5	400	49	4.34
C385	7	100	385	5	400	49	3.41
C283	6	100	283	5	400	49	2.61
Wrapping mesh:							
D98	5	200	98	5	200	98	1.54
D49	2.5	100	49	2.5	100	49	0.77
Stock sheet size	Length 4.8 m			Width 2.4 m			Sheet area 11.52 m ²
NOTE Values are subject to the tolerances specified in clause 10.							

7 Process of manufacture

Standard fabric (see annex B) shall be manufactured from wire of the same steel type and grade in both directions.

Purpose-made fabric (see annex B) shall be manufactured from wire of the same steel type and grade in each direction.

All fabric shall be factory made and machine welded. The joints at the intersections of the longitudinal wires and transverse wires shall be electrical resistance welded to provide shear resistant connections conforming to 13.2.

8 Quality

8.1 Only wire conforming to grades 460A or 460B of BS 4449 or to BS 4482 shall be used, except for wrapping fabric, with which wire of minimum yield strength 250 N/mm² shall be used (see Table 1).

8.2 Fabric including butt welded wires shall be permitted.

8.3 Fabric shall be free from any defect that can be shown to adversely affect the mechanical properties of the steel. Missing wires resulting from test sampling in accordance with annex A shall not be considered a defect.

8.4 The number of broken welds shall not exceed 4 % of the total number of cross welded joints in the sheet, nor exceed half the number of cross welded joints along any one wire.

9 Fabric classification

For reference and ordering purposes, the notation specified in BS 4466 for concrete reinforcement (see annex B) shall be used as a general basis for describing and classifying sheets of fabric.

10 Tolerances on mass, dimensions and pitch

10.1 Mass

The tolerance on the specified mass of the fabric per square metre shall be $\pm 6\%$.

10.2 Dimensions

The tolerance on the specified linear dimensions of the longitudinal and transverse wires in a sheet shall be ± 25 mm or 0.5 %, whichever is the greater.

10.3 Pitch

The deviation on the pitch of adjacent wires shall not exceed 15 mm or 7.5 % of the nominal pitch, whichever is the greater.

11 Bond classification of fabric

11.1 For bond purposes, fabric reinforcement conforming to this British Standard shall be welded steel fabric with shear resistant welded intersections conforming to 13.2, and with the number of welded intersections within its bond length conforming to BS 8110-1.

11.2 Welded steel fabric that does not conform to 11.1 with respect to the number of welded intersections within its bond length shall be classified for bond purposes as plain or deformed type 1 or deformed type 2 reinforcement (see BS 4482).

12 Routine inspection and testing

Routine inspection and testing shall be carried out as described in annex A.

13 Mechanical properties

13.1 Tensile and rebend tests

When tested in accordance with annex A, the tensile and rebend properties of the steel (except wrapping fabric) shall conform to BS 4449 or BS 4482, as appropriate. Wrapping fabric shall have a minimum yield strength of 250 N/mm².

13.2 Weld shear test

When tested in accordance with annex A, the shearing load required to produce failure of a welded intersection shall be not less than $0.25Af_y$, where A is the nominal cross sectional area of the small wire at the welded intersection and f_y is the specified characteristic yield stress of the wire.

14 Retests

If any sample fails the tensile, rebend or weld tests, two additional samples shall be taken from different sheets of the same batch and subjected to the failed test(s). If both additional samples pass the retests, the batch shall be deemed to conform to this British Standard. If either of the additional samples fails the retests, the batch shall not be deemed to conform to this British Standard.

15 Product identification

15.1 The steel shall be identified in accordance with BS 4449 or BS 4482 as appropriate.

15.2 Each bundle of welded fabric shall bear a label indicating the name of the manufacturer, the type of fabric, the number of this British Standard, and a batch traceability reference. If the fabric is covered by a third party product certification scheme, then this shall also be indicated on the label.

Annex A (normative)

Inspection and testing

A.1 Manufacturer's inspection

A.1.1 General

Test units shall be selected and tested in accordance with this annex.

A.1.2 Selection of test samples

Test units shall consist of sheets of a maximum mass of 50 t of the same type of fabric (or steels in the case of purpose-made fabric composed of different steel grades in the two directions) manufactured on the same welding machine.

Four test pieces (two for the longitudinal direction, two for the transverse direction) shall be taken from each test unit. Each test piece shall contain one or more cross welded joints.

NOTE The test pieces may be chosen from a single sheet or different sheets, provided that different wires are tested.

A.1.3 Condition of test specimens

Tensile tests shall be carried out on specimens in the aged condition.

Ageing shall be carried out as follows: heat the specimen to 100 °C and maintain it at this temperature for a minimum of 1 h. Cool to room temperature in still air.

A.1.4 Tensile test

Tensile strength, yield stress, total elongation at maximum force A_{gt} and/or elongation at fracture A_5 shall be tested in accordance with the appropriate material standard, i.e. BS 4449 or BS 4482.

A.1.5 Rebend test

Rebend testing shall be performed in accordance with the appropriate material standard, i.e. BS 4449 or BS 4482. The minimum distance between sections subjected to bending curvature and a welded intersection point shall be $4d$, where d is the nominal size of the test piece.

A.1.6 Weld strength test

Weld strength testing shall be performed in accordance with ISO 10287.

A.1.7 Test record

The manufacturer shall maintain records of the test results for 10 years from the date of testing. The records shall be available for inspection, on request, by the purchaser or purchaser's representative.

A.2 Material covered by a third party product certification scheme

A.2.1 Consistency of product

The long term quality level of the product shall be regularly assessed. (Long term quality level assessments shall not be used as a basis for conclusions regarding conformity to this British Standard.)

A.2.2 Determination of the long term quality level

A.2.2.1 Extent of determination

The following test results concerning all part quantities of continuous production shall be collated and statistically evaluated for yield strength (whichever is greatest in number). Either:

- all results from the preceding 3 months' production; or
- approximately 200 results.

A.2.2.2 Evaluation

Evaluation of results shall be performed for each diameter group (5 mm and 5.5 mm; 6 mm to 7.5 mm; 8 mm to 10.5 mm; 11 mm and over) and for each grade of steel.

Average values m [see A.3.4.1a)2)] shall conform to the following:

$$m \geq cv + k\sigma$$

where

- cv is the characteristic strength (N/mm²);
- k is the acceptability index (see Table A.1);
- σ is the standard deviation of the population.

Table A.1 — Acceptability index k as a function of the number of test results n for a reliable failure rate of 5 % (pass = 0.95) at a probability of 90 % ($1 - \alpha = 0.90$)

Number of test results n	Acceptability index k
5	3.40
6	3.09
7	2.89
8	2.75
9	2.65
10	2.57
11	2.50
12	2.45
13	2.40
14	2.36
15	2.33
16	2.30
17	2.27
18	2.25
19	2.23
20	2.21
30	2.08
40	2.01
50	1.97
60	1.93
70	1.90
80	1.89
90	1.87
100	1.86
150	1.82
200	1.79
250	1.78
300	1.77
400	1.75
500	1.74
1000	1.71
∞	1.64

A.2.3 Certificate of conformity

A certificate of conformity shall be issued if requested by the purchaser, stating:

- that the material has been tested in accordance with this British Standard and at the specified frequency;
- the address where the test results are kept available for inspection.

The certificate shall bear the approval number issued by the certifying authority.

A.3 Material not covered by a third party product certification scheme

A.3.1 General

Material not covered by a third party product certification scheme shall be assessed by acceptance tests on each batch. Sampling shall be carried out by an independent inspection authority at the producer's works or in the stockholder's yard, with subsequent testing by an independent accredited test laboratory.

A.3.2 Extent of sampling and testing

Test units shall have a maximum mass of 50 t and shall consist of the same type of fabric and the same grade of steel. Certificates shall be issued giving the chemical composition of the materials and stating that all products in the test unit originate from the same production batch.

Test specimens shall be taken from each test unit as follows:

- for testing in accordance with A.3.3, 20 test specimens [or 80 specimens if appropriate; see A.3.4.1b)], from different sheets of fabric;
- for testing chemical composition, four test specimens, from different sheets of fabric.

Test specimens shall be prepared in accordance with A.1.

A.3.3 Tests

Specimens selected in accordance with A.3.2 shall be subjected to the following tests.

- Inspection of variables*
 - Tensile strength (R_m).
 - Yield strength (R_e).
 - Ratio R_m/R_e .
 - Elongation at fracture (A_5).
 - Total elongation at maximum force (A_{gt}) (if applicable).
 - Weld shear strength.
- Inspection by attributes*
 - Response to the rebend test.
 - Deviation from the nominal cross sectional area.
 - Surface geometry (if applicable).
- Chemical composition in accordance with product analysis*

All elements listed in the appropriate material standards, i.e. BS 4449 or BS 4482.

A.3.4 Evaluation of results

A.3.4.1 Inspection by variables

Inspection by variables shall be performed as follows.

a) Determine the following indices for the properties listed in **A.3.3a**):

- 1) individual values x for the 20 test specimens ($n = 20$);
- 2) the mean value m_{20} (for $n = 20$);
- 3) the standard deviation S_{20} (for $n = 20$).

The test unit shall be deemed to conform to this British Standard if the following condition applies:

$$m_{20} - 2.21S_{20} > cv$$

where

2.21 is the acceptability index k ;

cv is the characteristic strength specified in the appropriate material standard, i.e. BS 4449 or BS 4482.

b) If the condition stated in **A.3.4.1a**) does not apply, determine the alternative acceptability index k' as follows:

$$k' = \frac{m_{20} - cv}{S_{20}}$$

If $k' > 2$, testing shall be continued. Sixty further test specimens shall be selected from different sheets and tested, making a total of 80 test results available ($n = 80$).

The test unit shall be deemed to conform to this British Standard if the following condition applies:

$$m_{80} - 1.89S_{80} > cv$$

where

1.89 is the acceptability index k for $n = 80$.

c) The ratio R_m/R_e , elongation at fracture A_5 and total elongation at maximum force A_{gt} shall conform to the appropriate material standard, i.e. BS 4449 or BS 4482. Weld strength shall conform to **13.2**.

A.3.4.2 Inspection by attributes

Either:

- a) if the results of tests on the properties listed in **A.3.3b**) conform to the appropriate material standard, i.e. BS 4449 or BS 4482, for all 20 test specimens, the test unit shall be deemed to conform to this British Standard; or
- b) if a maximum of two of the 20 results do not conform to a), 80 further test specimens shall be tested from different sheets in the test unit. The test unit shall be deemed to conform to this British Standard if no more than two of the 80 test results fail to conform to the appropriate material standard, i.e. BS 4449 or BS 4482.

A.3.5 Test report

All fabric subjected to batch testing shall be accompanied by a test report, specifically related to the batch, endorsed by the inspection authority (see **A.3**) and containing the following information:

- a) the name of the manufacturer;
- b) the place of manufacture of the fabric;
- c) the type of fabric;
- d) the grade of reinforcing steel;
- e) the date of testing;
- f) the mass of the test unit;
- g) batch numbers;
- h) results of tests in accordance with **A.3.3**;
- i) an official stamp of the inspection authority;
- j) marking on the reinforcing steel (if appropriate).

Annex B (informative)

Notation and classification of fabric

B.1 General

Figure B.1 gives an example of notation that should be used for ordering sheets of fabric described and classified as specified in BS 4466 (see clause 9).

Steel grades should be designated as specified in BS 4466.

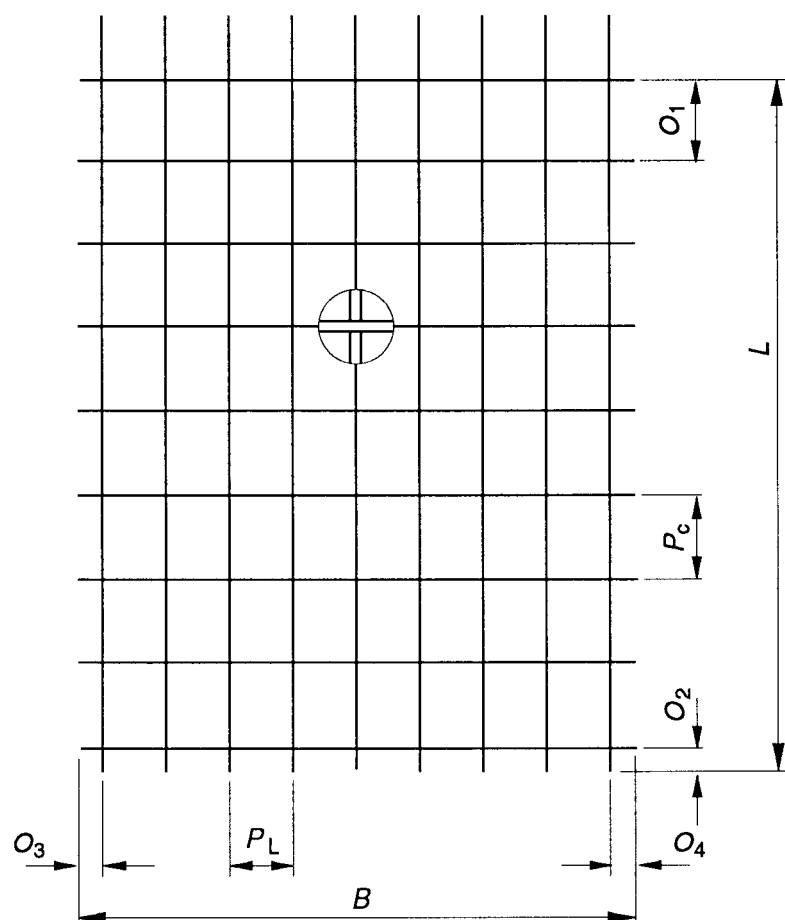
For standard and detailed fabric (see BS 4466), a simplified procedure may be adopted (see **B.2** and **B.3**).

B.2 Standard fabric

Standard fabric has defined mesh arrangements and wire sizes and should be specified by designated reference numbers. BS 4466 gives an example of a standard fabric from the preferred range (see Table 1), with an alphanumeric reference to define the size and pitch of the longitudinal wires and cross wires.

B.3 Purpose-made fabric

When the standard form of schedule is used to define more complicated arrangements, involving longitudinal wires of different sizes or irregular spacings in one or both directions, a fully dimensioned diagram should supplement the schedule (see for example Figure B.2).

**Key**

- L length of longitudinal wires (not necessarily the longest wires in the sheet)
- B length of cross wires
- O_1, O_2 overhangs of longitudinal wires
- O_3, O_4 overhangs of cross wires
- P_L pitch of longitudinal wires
- P_C pitch of cross wires

Figure B.1 — Fabric notation

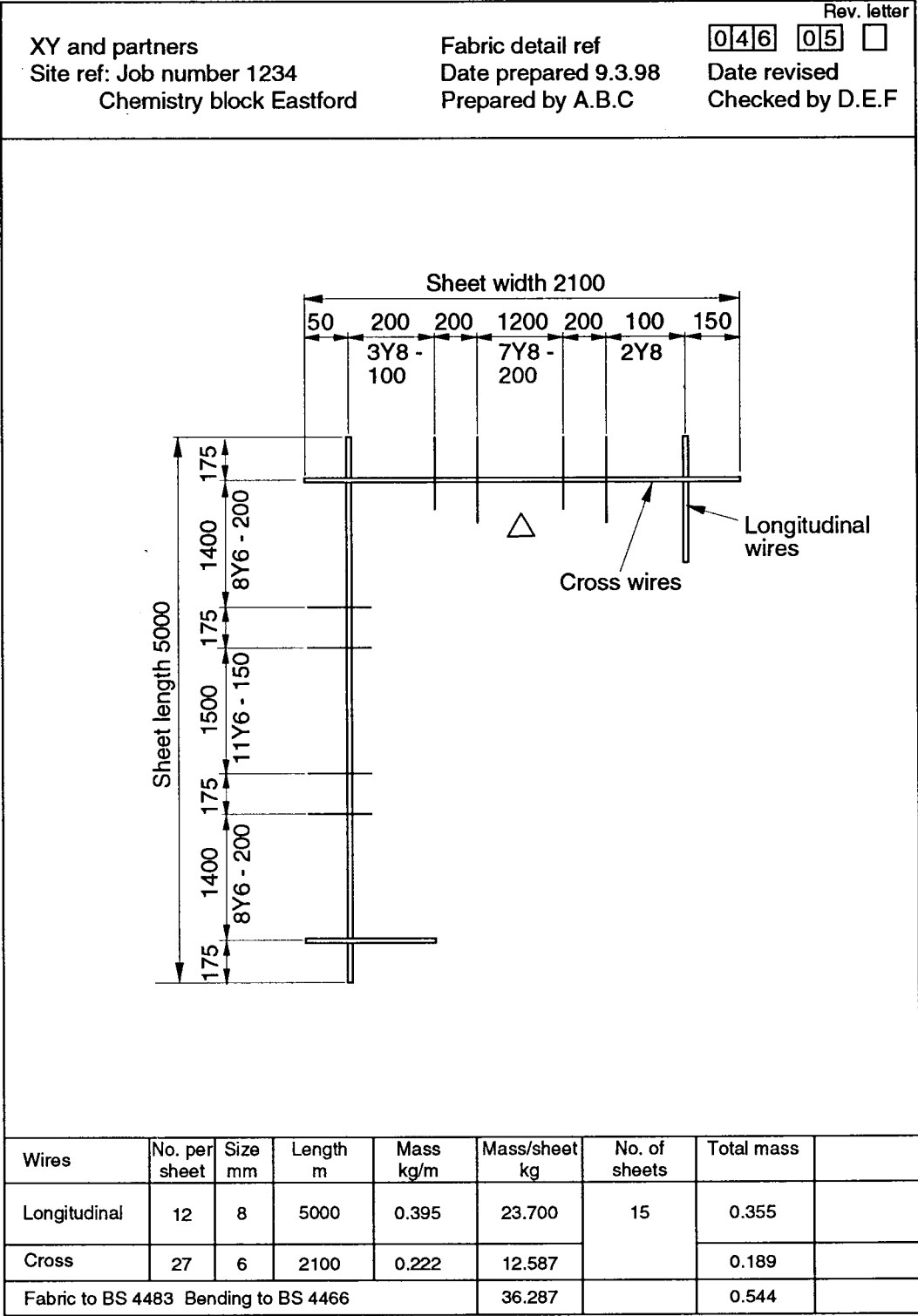


Figure B.2 — Purpose-made fabric example

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